higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CARPENTRY AND ROOF WORK
NQF LEVEL 4

22 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 1.1.1 A platform built from wood√ to support a geyser in a roof.√

1.1.2 The horizontal beam forming the base of a triangular truss in a pitched roof, √ connecting the two side walls and preventing the two rafters from separating.√

1.1.3 A leak-proof structure made from plastic or steel plate√ designed to collect any leak or overflow of water from the geyser.√

1.1.4 A beam√ supporting a floor or ceiling.√

1.1.5 Truss members that connect and support√ the bottom and top chords.√

(5 × 1) (5)

1.2 The nailing method used for fixing the additional member is termed 'staggered nailing'✓ and this prevents splitting of the additional member.✓ The timber used for the support must be the same grade as the existing timber✓ and the additional member should extend past every node by not less than 150 mm.✓ The additional member must be at least 2.5 meters long✓ and a minimum of three trusses is required for the support deck.✓ A competent person must be assigned for the design of the support structure. A structural engineer must design a support structure for geyser capacity more than 250 litres and a carpenter can design and install support decks for geysers smaller than 250 litres.✓ (8)

1.3
- The truss is weakened.
- The truss can snap. (2)

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QUESTION 2

2.1
- Aesthetics:✓ A ceiling would hide the inner structure of a roof construction and moulded or painted ceilings adds to the visual appeal of a room.✓
- Thermal comfort:✓ Ceilings help to limit temperature changes.✓
- Strength:✓ Prefabricated buildings with lightweight wall panels are strengthened with ceilings that will reduce movement.✓
- Sound insulation:✓ Ceilings reduce the transfer of noise e.g. music from one room to another.✓
- Dust:✓ Ceilings prevent dust that is blown under the roof from outside.✓
- Fire:✓ Ceilings that are made from fire-proof material will prevent the spread of fire.✓ (12)

2.2 450 mm✓ otherwise the gypsum board might sag. ✓ (2)

2.3 Packing strips are placed in the space between the bottom chord and brandering✓ to level the ceiling.✓ (2)

2.4
- It improves the appearance of the ceiling in a room.
- It conceals the gap where the ceiling meets the wall.
- It prevents dust from the roof entering the room. (3)

2.5 760 mm✓ × 760 mm✓ (2)

2.6 100 kg (1)

2.7 2.7.1 The trap door cannot be installed in the position because it is too close to the eaves✓ of the house that will make it difficult to get into the room. It is also to far from the geyser and on top of a kitchen cupboard that will make it difficult to enter.✓ (Any applicable 2 × 1) (2)

2.7.2 The trap door cannot be installed because it is too close to the eaves✓ of the house and in the middle of the Dining room that will affect the appearance of the ceiling. It is also to far from the geyser.✓ (Any applicable 2 × 1) (2)

2.7.3 The trap door can be installed in this position because it is close to the geyser✓ and in the middle of the roof structure that will give enough space to enter into the roof. It is also hidden from the main rooms where it is less visible.✓ (Any applicable 2 × 1) (2)

2.8
- Shredded paper treated with borax
- Fibreglass
- Soft polyester (3)

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QUESTION 3

3.1
- It adds to the aesthetic value of the building and look very attractive in comparison with the cold appearance of the concrete floors that need additional material to make it more attractive e.g. tiles.
- Timber floors are light in weight causing smaller load to the structure of the building in comparison with heavy concrete slabs.
- Concrete floors are cold in winter whereas wooden floors are good heat insulators.
- The Suspended timber ground floors are more expensive due to the construction material in comparison with concrete slab casted on damp-proof membranes.
- The labour cost to construct a suspended timber ground floor is more and is more time consuming in comparison with concrete floors that can be casted in a very short time.
- The concrete floor is rot-proof and will last very long in comparison with the wooden construction of the suspended timber floor that must be treated against fungi and insects that can damage the wooden construction.

- Suspended timber floors are warmer and have good heat insulation in comparison with concrete floors that are cold in the winter.
- Concrete floors can carry more weight than timber floors and is usable for construction purposes early in the construction process. (8)

3.2
- Laminated beams resist expansion and contraction in different weather conditions.
- Laminated beams are less prone to insects and fungus attacks.
- Laminated beams can span longer bearing distances. (3)

3.3
- Good ventilation will prevent dampness underneath the floor.
- Good ventilation will prevent the flooring boards from warping.
- Good ventilation can prevent the floor from rotting. (3)

3.4 3.4.1 A short wall✓ used for central support of floor joists✓ (2)

3.4.2 A metal barrier built into the foundation masonry✓ that stops ants and other termites from crawling through the building✓ (2)

3.4.3 A horizontal barrier in a wall✓ that prevents moisture from rising✓ (2)
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QUESTION 4

4.1 4.1.1 True (1)

4.1.2 False: A doorframe should be in a shape of a rectangle with all corners 90°. (2)

4.1.3 False: If the jambs of a doorframe is slightly misaligned, you could tweak it by moving the hinge on the contact side of the door away from the rebate by a maximum of 3mm. (2)

4.1.4 False: If the floor is not level and runs out of level downwards, away from the door, you do not need to worry when fitting and hanging a door. (2)

4.1.5 True (1)

4.2
- Assess the doorframe and door
- Mark the doorframe and the door simultaneously
- Fit the door
- Hang the door
- Ease the door
- Install the lock and other door furniture (6 × ½) (3)

4.3
- Determine if it is a left or right-hand lock that you need to install and make the necessary adjustments.
- Measure and mark the position of the lock and spindle hole on the side of the door where the lock will be installed.
- Mark the position of the lock on the edge of the door.
- Drill a series of holes on the edge of the door where the lock will slide in.
- Clean the slot with a chisel until the lock will slide in.
- Cut a recess where the lock will be seated into the edge of the door. The face of the lock must be flush with the door.
- Drill the holes for the spindle and key holes with a 16 mm electric drill.
- Fit all parts and insert all the screws.
- Test the lock and make sure it can open and close and lock. (9)

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QUESTION 5

5.1
- Manufactured boards do not need too much preparation and finishing.
- Manufactured boards can span greater distances than solid timber.
- Manufactured boards are cheaper than solid timber. (3)

5.2 The plinth is the base on which the cupboard stand✓ and should be high enough to clear the skirting.✓ The kickboard that is the same material as the cupboards will be fixed to the plinth.✓ The plinth should be made of solid wood for example pine because the chipboard swell and lose its properties with the slightest bit of moisture.✓ (4)

5.3 A profile gauge enables the carpenter to copy and transfer irregular shapes easily.✓The gauge is used to copy the profile of the surface to which the trim must be fitted.✓ The profile is then used to mark the profile onto the trim material to be cut.✓ The profile gauge consists of steel teeth that are set tightly against each other and when it is pressed against an object, the teeth will take the shape of the object's profile.✓ (4)

5.4
- During transportation the handles can damage other cabinets and break off.
- The handles must be installed only after the cupboard doors have been aligned. (2)

5.5
- The contents of the shelves accumulate dust.
- Additional central support is needed to prevent sagging. (2)

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TOTAL: 100

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